THE PENNSYLVANIA RAILROAD

Locomotive Maintenance Instructions No. L-5-D

ISSUED ALTOONA, PA.
JULY 17, 1933

Method of Applying and Maintaining Flues and Arch Pipes in Locomotive, Stationary and Portable Boilers and Instructions for Welding Flues to Flue Sheets

(SUPERSEDING LOCOMOTIVE MAINTENANCE INSTRUCTIONS NO. L-5-C, DATED DECEMBER 13, 1923)

GENERAL

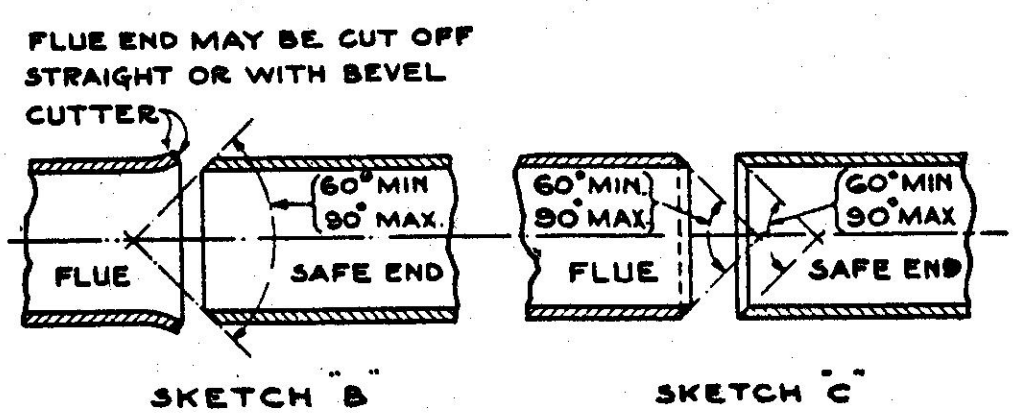
1. **Flues and Arch Pipes:** Boiler flues and arch pipes must be in accordance with Standard Specification No. 17 and the tracings designated. Where specifications, instructions or tracings are referred to, the latest issue of the same is intended. Boiler flues and arch pipes should be scrapped when worn two guages less than thickness shown on standard tracing.

2. **Safe Ends:** (a) Safe ending material must be .134 inch in thickness, B.W.G. No. 10, for repairing all flues that have a wall thickness of .134 or less. For repairing all flues that have a wall thickness of .148 inch, B.W.G. No. 9, or over, the thickness of the safe ending material must be the same as that of the original flue repaired.

(b) The outside diameter of safe ending material must be the same as that of the flue end to which it will be applied, except the material used for safe ending the 3½″ and 5½″ superheater flues, which may be 1/16″ larger in diameter to avoid the necessity of opening the end up to suit holes in front flue sheet.

(c) Where possible, the safe ends should be applied to the firebox end of the flues, except the large superheater flues 3½″ and 5½″ outside diameter, which must have the safe ends applied to the smokebox end of the flue. The length of the safe ends must not be less than 3½″ measured from the outside of flue sheet after the flues are applied, for the small flues, and 4½″ for the superheater flues. Arch pipes and water tubes must not be safe ended.

(d) The weld end of flues and safe ends that are to be welded by the oil fired method must be prepared as shown in sketch "B". The weld end of flues and safe ends that are to be welded by the electric resistance method must be prepared as shown in sketch "C". When the wall thickness of old flues is found to be less than two gauges below that specified on standard tracings for new flues, it must be scrapped. The flues must be tested under a water pressure of not less than 300 pounds and all the welds that are tested by the compression test method must be rapped while under pressure with a hammer for defects. Flues that are tested with a tension flue tester while under pressure need not have the welds rapped with a hammer. The inspector must then stamp on the safe end, or ends, the designated mark assigned the welder for identification in cases of failure in service.



(e) Bend tests for safe end welds must be made for each welder. A sample being taken for each four hours work. These tests are to be made on both superheater and small flues and the results obtained must be submitted to the Master Mechanic for approval. The sample for test must be a portion of approximately 2″ wide by 5″ in length cut from the welded flue in such a manner as to bring the safe end weld in the